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MODEL EVALUATION

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Dataset : glass.csv

Shape : (214, 10)

Binary Class Distribution

Type

0 144

1 70

Name: count, dtype: int64

Training Samples : 149

Testing Samples : 65

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CONFUSION MATRIX

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[[36 8]

[9 12]]

True Positive : 12

True Negative : 36

False Positive: 8

False Negative: 9

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PERFORMANCE METRICS

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Accuracy : 0.7385

Precision: 0.6000

Recall : 0.5714

F1 Score : 0.5854

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DERIVED METRICS

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True Positive Rate (TPR): 0.5714

True Negative Rate (TNR): 0.8182
False Positive Rate(FPR): 0.1818
False Negative Rate(FNR): 0.4286

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ROC CURVE

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AUC Score : 0.8290

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SUMMARY

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Accuracy : 0.7385
Precision: 0.6000
Recall : 0.5714
F1 Score : 0.5854
AUC : 0.8290

Model evaluation completed successfully.

The Logistic Regression model achieved an Accuracy of 73.85% and an AUC score of 82.90% on the Glass Identification dataset. Although precision and recall indicate moderate classification performance, the ROC curve demonstrates that the model can distinguish between positive and negative classes reasonably well.

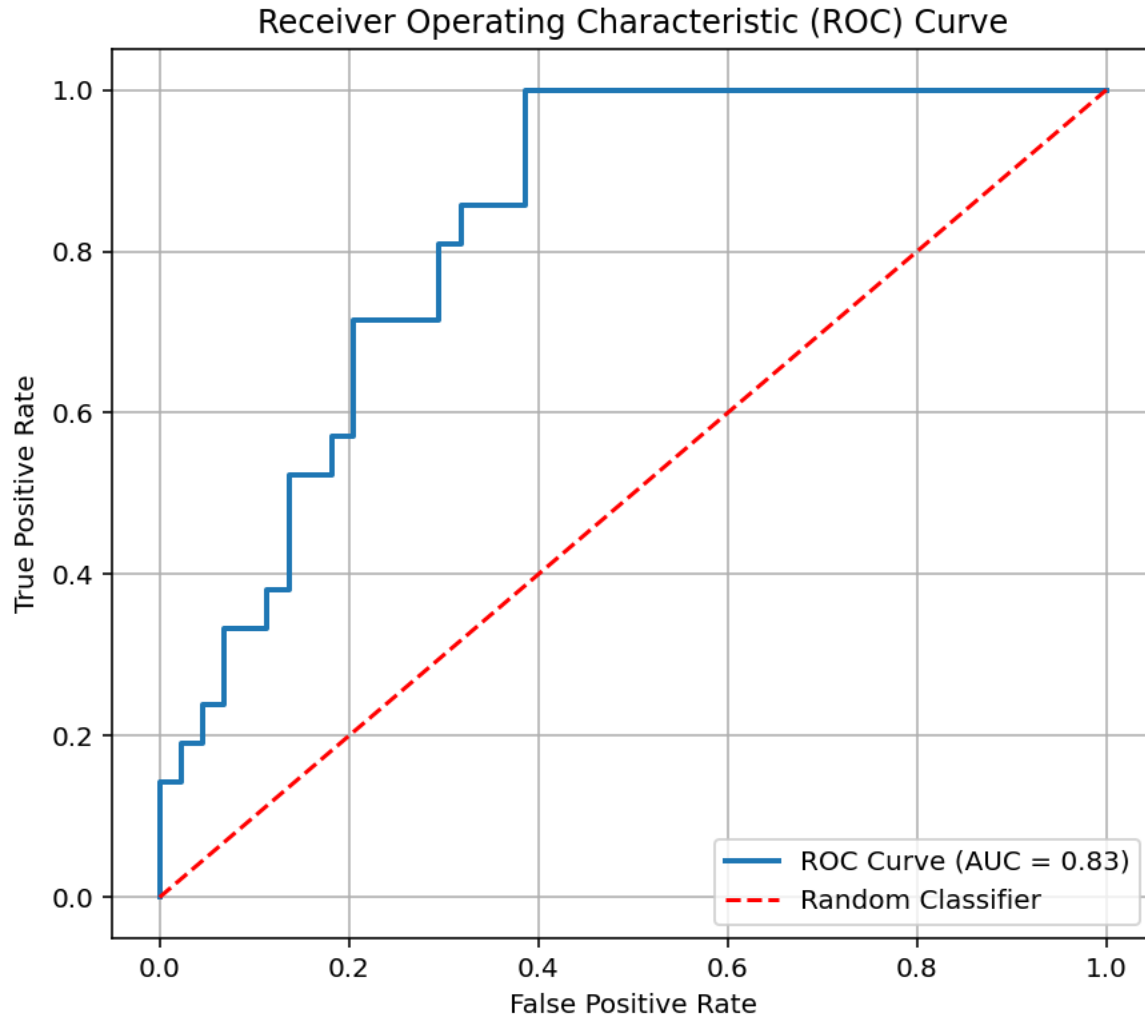


Figure 1. Receiver Operating Characteristic (ROC) Curve for Logistic Regression on the Glass Identification Dataset

The Receiver Operating Characteristic (ROC) curve evaluates the performance of the Logistic Regression classifier on the Glass Identification dataset after converting the multiclass problem into a binary classification task. The curve illustrates the relationship between the True Positive Rate (Sensitivity) and the False Positive Rate at different decision thresholds.

The blue ROC curve remains well above the red diagonal reference line, which represents the performance of a random classifier. This indicates that the model is able to distinguish between the positive and negative classes more effectively than random guessing.

The classifier achieved an Area Under the Curve (AUC) score of 0.83, indicating good classification performance. An AUC value closer to 1.0 represents a better classifier, while an AUC of 0.5 corresponds to random prediction. Therefore, an AUC of 0.83 demonstrates that the trained model has a strong ability to correctly classify unseen samples.

